

1)

a)

The origin of paramagnetism and diamagnetism comes from when inspecting a magnetic material on the atomic scale, you would see small currents around each nuclei like small current loops creating magnetic dipoles. These will normally cancel each other out as they all point in random directions, but when an external B-field is applied, a net alignment of the dipoles occurs turning the material magnetically polarized/magnetized. The direction of magnetization is what determines if its paramagnetic, M is parallel to B, or if its diamagnetic, M is opposite to B.

b)

Ferromagnetic materials can remain strongly magnetized after an external magnetic field because each dipole in the material "likes" to point in the same direction as the other closest dipoles. However this alignment occurs in domains with billions of dipoles that may not align with another domain. When an external field is applied, the magnetic torque at boundary between two domains will be able to turn some of the dipoles toward alignment of the B-field, turning the material saturated if the field is strong enough. This quality is not usually present in paramagnetic materials, which is why they don't stay magnetized.

c)

For ferromagnets, the magnet dipole moment is generally more aligned than in para- & diamagnets, leading to a much larger magnitude in the total magnetization thus stronger effects.

d)

Material	Para	Dia	Explanation
Water		x	Paired electrons, creates opposing magnetic field
Magnesium	x		Unpaired electrons, creates aligning magnetic field
Silver		x	Paired electrons, creates opposing magnetic field
Carbon	x		Unpaired electrons, creates aligning magnetic field
Sodium Chloride		x	Paired electrons, creates opposing magnetic field
Molecular oxygen	x		Unpaired electrons, creates aligning magnetic field

Paramagnetic vs Diamagnetic

Paramagnetic



s orbital



p orbital

- Unpaired electrons
- Attracted to a magnetic field







Diamagnetic

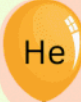


s orbital



p orbital

- Paired electrons
- Repelled by a magnetic field







All materials are diamagnetic, but in atoms with unpaired electrons, paramagnetism overcomes diamagnetism.

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2)

a)

The loop will experience torque due to the misalignment of the magnetic dipole m and the magnetic field B , however the induced forces on the sides of the loop will point in opposing directions on each side. This is because the current flows opposite direction on each matching side, thus cancelling out the forces (no net force).

b)

The torque tends to align the orientation of the magnetic dipole towards what is closest of parallel or opposite to the magnetic field depending on θ . From the right hand rule we can get the direction of how it will rotate, for this case looking at the yz -plane, $m \times B$ leads to "the thumb" going out of the page (x -axis), and the torque will then rotate it along this axis in the direction of the positive z -axis since θ is < 90 degrees.

c)

The net forces on the sides of the loop will cancel when looked at mathematically/in translation, but they will act on different points of the loop leading to the rotational torque. This makes it so that the non-aligned loop will have higher potential energy than the aligned loop, and the torque will use that energy as the work to rotate it back.

d)

The key similarity is that the torque for both is the dipole moment $\times$ the E- or B-field for uniform fields. One important physical difference is that for E-fields this works for static charges, while for B-fields its for steady current loops. Using Gilbert model of a magnetic dipole (seperate monopoles) thats similar to electric dipole, is physically incorrect on a smaller scale of analyzation, but it can be helpfull to get a "feel" for a problem.

3)

a)

If the bound material as bound currents J_b and K_b , as with them you can calculate M and then use that:

$$\oint H \cdot dl = I_{f_{enc}} \implies H = 0$$

$$H = \frac{1}{\mu_0} B - M \implies B = \mu_0 M$$

b)

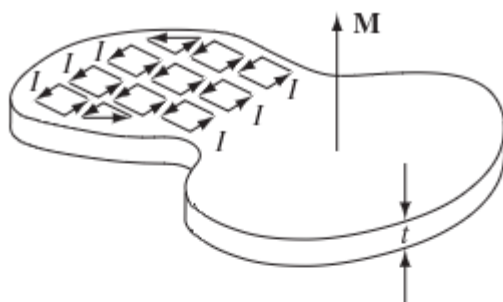


Fig. 6.15

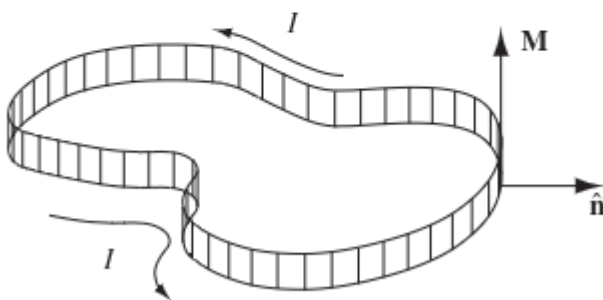


Fig. 6.16

As seen on the figure 6.15, we can represent the dipoles of a material with magnetization as small current loops. If it is uniform, the internal current loops will cancel each other, and the bound internal current density $J_b = 0$. However if its

nonuniform, then it can be proved that $J_b = \nabla \times M$ which means that the internal bound current density of a nonuniform magnetic material is the curl or rotation of the magnetization M . This comes from calculating the vector potential of a single dipole m , and looking at the part representing the potential of a volume current.

c)

The bound surface current density can be seen again from figure 6.15 as for a uniform magnetized material, the internal currents will cancel out, while the currents along the edge will not. This leads to the entire area's current can be described as the current running through the surface like in figure 6.16. The total surface current density will then be equal to the magnitude of the magnetization, and using the $\hat{n}$ vector we can get the direction as $K_b = M \times \hat{n}$. The current is bound as it consists of small loops of currents that have a small net current around the material.

d)

As explained earlier, the internal currents in a uniformly magnetized object will cancel each other out by the "neighbouring" currents, while the currents along the surface will not as they have no "neighbour".

e)

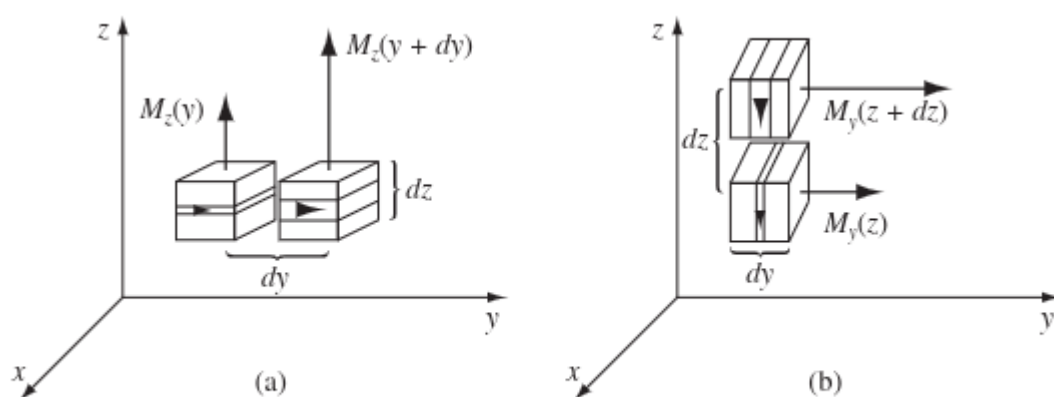


Fig. 6.18

When a material is nonuniformly magnetized, the internal currents will vary in magnitude and orientation due to varying magnetization. This in turn leads to the internal currents between the current loops, as seen in Figure 6.18, will not cancel each other due to one being larger.

4)

4) a)



$$s < a: \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} = \underline{\underline{0}}$$

$$a < s < b: \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

$$\Rightarrow 2\pi s |\mathbf{H}| = \frac{\pi (s^2 - a^2)}{\pi (b^2 - a^2)} I$$

$$\Rightarrow \underline{\underline{\mathbf{H} = \frac{I}{2\pi s} \frac{(s^2 - a^2)}{(b^2 - a^2)} \hat{\varphi}}}$$

$$s > b: \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \Rightarrow 2\pi s |\mathbf{H}| = I$$

$$\Rightarrow \underline{\underline{\mathbf{H} = \frac{I}{2\pi s} \hat{\varphi}}}$$

b) $s > b: \mathbf{M} = 0 \Rightarrow \mathbf{B} = \mu_0 \mathbf{H} \Rightarrow \mathbf{H}$ same direction as $\mathbf{B}$ against the clock.

$s < a: \mathbf{M} = \mathbf{H} = 0$ No direction

$a < s < b: \mathbf{M}$ is not known, but $\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} + \mathbf{M}$

since we know that $\mathbf{M}$ is opposite of $\mathbf{B}$ (diamagnetic) and thus $\mathbf{H}$ is same direction as $\mathbf{B}$, against the clock.

c) $\mathbf{M}$ does not change the calculations of $\mathbf{H}$ as it's related to the bound currents, and $\mathbf{H}$ is related to the free currents.